

On the Unexpected Reliability of Coffee Cups as Fiducial Markers in Indoor Video Sequences

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Abstract—We investigate whether ordinary coffee cups can support repeatable localization in indoor video. A synthetic, fully controlled study compares cup rims with printed markers under occlusion, glare, and deliberate desk rearrangement. The cup is useful when its rim remains visible, although its handle introduces a direction-dependent bias. The study illustrates both the promise and the limits of conveniently placed objects.

Keywords—video analysis; temporal reasoning; reproducibility; synthetic experiments.

1 Introduction

A calibration marker is helpful until nobody remembers to place it in the scene. Coffee cups, by contrast, tend to arrive without a protocol. We study whether that abundance can be turned into a modest geometric cue. Our aim is localization within a desk scene, not metric camera calibration.

2 Related Work

Fictional studies of incidental landmarks emphasize stable contours and repeated observations [1, 2]. Work on beverage-vessel tracking describes the handle as both a distinctive feature and a source of view-dependent error [3]. We isolate the rim so that this ambiguity can be measured.

3 Methodology

We detect candidate ellipses in grayscale frames, reject contours with low arc coverage, and associate candidates across time by nearest-neighbor matching. A candidate receives confidence $q = 0.5c + 0.3s + 0.2t$, where c is visible rim fraction, s is shape consistency, and t is temporal persistence. The score is descriptive and was fixed before evaluation.

4 Experimental Setup

We rendered 120 short desk sequences at 1280×720 pixels. Each contains one cup, one sheet of paper, and a varying number of pens. Camera motion, glare, and occlusion were independently varied. Ground-truth cup centers were exported

by the renderer; no person was recorded.

5 Failure Model

An ellipse fitted to a short arc may be geometrically plausible while its center is poorly determined. We therefore retain the angular coverage of the observed rim, the residual of the fit, and the distance to the preceding center. A candidate is rejected if its center jumps by more than 35 pixels without corresponding camera motion. This deliberately conservative gate favors continuity over recovery after a long occlusion.

The printed-marker comparator uses four corners and the same association window. Both pipelines receive identical frames and are evaluated only when their respective object is present. We distinguish a missed detection from a center error: assigning a small localization error to a lucky but intermittent detection would conceal the practical cost of missing frames.

6 Evaluation Protocol

Sequences are grouped by rendered desk layout before computing the summary. We use the median error for successfully localized frames and separately report coverage, the fraction of frames with an accepted estimate. Thresholds are chosen on 20 development sequences, leaving 100 for the reported evaluation. No frame from a development layout appears in the evaluation set. The renderer records the cup center before glare and blur are applied, giving an unambiguous target but an optimistic measure of labeling precision.

7 Results

At full visibility, median center error was 2.8 pixels; under partial occlusion it rose to 8.6 pixels. A printed marker achieved 1.4 and 5.1 pixels, respectively. Cup failures clustered at specular rims and when the visible arc fell below one half. The method remained serviceable as a rough reference, but was unsuitable for precise calibration.

Cue	Clear	Occluded
Printed marker	1.4	5.1
Cup rim	2.8	8.6
<i>Median center error (pixels).</i>		

8 Discussion

The synthetic scenes are deliberately simple. Real cups vary in shape and are often moved between frames, sometimes by the very person being observed. The score also cannot distinguish a cup from a similarly shaped bowl. These constraints make the cue most useful for exploratory tracking.

9 Additional Analysis

Clear-view coverage was 96% for the cup cue and 99% for the printed marker. Under occlusion those values fell to 72% and 88%. The center-error comparison therefore describes only accepted estimates; the lower cup coverage is a separate limitation. Raising the confidence threshold improved median cup error by roughly one pixel but discarded another 11% of frames. This tradeoff matters when a downstream system needs continuous trajectories.

A qualitative inspection identified three recurring cases. First, overhead light produced a bright arc outside the actual rim. Second, a handle joined the rim contour and shifted the fitted ellipse. Third, a white saucer produced a second plausible ellipse below the cup. Temporal matching usually repaired the first case, but not a sustained saucer confusion. We include these cases as named failure modes rather than folding them into a single aggregate score.

10 Practical Implications

The estimator can initialize a region of interest for later tracking when a cup is already part of the scene. It should not be used to infer camera pose or physical distance without a known cup diameter and a separate calibration. For a reproducible real-world follow-up, cup dimensions, lighting position, and the fraction of visible rim should be recorded with each clip. A useful test would also deliberately remove the cup mid-sequence to measure how quickly the tracker recognizes that its reference has disappeared.

11 Reproducibility Notes

All synthetic settings are fixed at the sequence level rather than changed within a clip. A future release should include the generation seed, object meshes, and the exact contour thresholds alongside the frames. Reporting only the successful localization error would make a sparse estimator look better than a stable one; coverage and failed-frame examples should travel with the summary table. The development split also needs to remain separate when new cup shapes are introduced.

12 Conclusions and Future Work

A coffee cup can serve as an incidental, low-precision landmark when its rim is visible. Future work will test real cameras, more vessel shapes, and detection confidence calibrated against actual localization error.

References

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